Answer Key

Grade 6 Mathematics · Unit 4: Working with Percents · Answers for every item on Sheets 1 and 2 of each lesson packet (core and Level 1 have the same answers). Constructed responses show a model answer; accept equivalent reasoning.

LESSON 4-1

Percent Means Per Hundred · Percents Greater Than 100%

Sheet 1 · Percent Means Per Hundred

1. a. 46% b. 54%
2. a. 30% b. 28% c. 50% d. 34% e. 40% f. 15% g. 52% h. 90%
3. Missing: 27, 36. 9 out of 25 = 36 out of 100 = 36%.
4. a. 22 × 2 = 44 out of 100 = 44% b. 100% − 44% = 56%
5. Equivalent ratios are made by multiplying (or dividing) both terms by the same number, not adding. 20 × 5 = 100, so 7 × 5 = 35: 7 out of 20 = 35%.

Sheet 2 · Percents Greater Than 100%

1. a. 300% b. 175%
2. Missing percents: 50%, 75%, 125%, 150%, 175%. Tuesday = 175% of Monday.
3. 260 to 200 = 130 to 100, so 130% (more than the goal).
4. 33 to 20 = 165 to 100 (multiply both by 5), so 165%.
5. 60% of 30 cm is only 18 cm, which is less than 30 cm. 60 is 2 times 30, so 60 cm is 200% of 30 cm.

TEST PRACTICE

6. **D** (150%)
7. **B** (2.5 times)

LESSON 4-2

Fractions, Decimals, Percents · Compare and Order

Sheet 1 · Fractions, Decimals, Percents

1. a. 35/100 = 7/20 b. 0.35 c. 35%
2. 1/4 = 0.25 = 25%; 7/10 = 0.7 = 70%; 2/25 = 0.08 = 8%; 13/20 = 0.65 = 65%; 3/5 = 0.6 = 60%
3. 11/25 = 44/100 = 0.44 = 44%
4. She just wrote the digits 3 and 4 side by side; that is not the same value. 3/4 = 75/100 = 0.75 = 75%.

Sheet 2 · Compare and Order

1. a. > b. < c. <
2. a. 7/10, 0.72, 75% b. 0.2, 22%, 1/4 c. 6/5, 125%, 1.3
3. A = 55%, B = 60%, C = 52%. Order: Room C, Room A, Room B (Room B greatest).
4. 3/5 = 60% and 0.7 = 70%, so any value between 60% and 70%. Sample: 13/20 = 0.65 = 65%.
5. Comparing single digits does not compare values; change to one form. 3/8 = 0.375 = 37.5% and 0.4 = 40%, so 3/8 < 0.4.

TEST PRACTICE

6. **C** (0.35, 38%, 2/5)
7. **C** (64%)

LESSON 4-3

Estimate with Benchmarks · Estimate a Percent of a Number

Sheet 1 · Estimate with Benchmarks

1. Pages: 40, 80, 120. 42 is close to 40, so about 25%.
2. a. 10% b. 75% c. 25% d. 50% e. 100% f. 25% g. 10% h. 75%
3. a. $50 b. 20% of $500 = 2 × $50 = $100, and $96 is close to $100, so about 20%. Also accept about 25% with reasoning ($96 is between 10% = $50 and 25% = $125, and closer to $125).
4. 75% of 48 = 36, and 35 is close to 36, so about 75%.
5. 26 out of 50: half of 50 is 25, so it is just over 50% (exactly 52%). 26 out of 60: half of 60 is 30, so it is below 50% (about 43%). 26 out of 50 is closer to 50%.

Sheet 2 · Estimate a Percent of a Number

1. a. about 6 (10% of 60) b. about 30 (75% of 40) c. about 50 (25% of 200)
2. a. $20 b. about $60 (3 × $20)
3. 50% of 400 = 200, so about 200 students.
4. 25% of 800 = 800 ÷ 4 = 200, so about 200 apples.
5. 20% is not 20; she multiplied by 20 instead of 0.20. 10% of 500 = 50, so 20% of 500 = 100. 19% of 512 is about 100.

TEST PRACTICE

6. **B** (20)
7. **C** ($48)

LESSON 4-4

Find the Percent of a Number · Compare Percents

Sheet 1 · Find the Percent of a Number

1. a. 9 b. 48 c. 21 d. 36 e. 45 f. 30
2. Missing: 72, 108, 144. 40% of 180 = 72 students.
3. a. 0.25 × 64 = $16 b. $64 − $16 = $48
4. 10% of 40 = 4 and 5% = 2, so 15% = 6 players (0.15 × 40 = 6).
5. They are equal: 0.3 × 90 = 27 and 0.9 × 30 = 27. Both equal 30 × 90 ÷ 100, so switching the numbers gives the same amount.

Sheet 2 · Compare Percents

1. a. 70% b. 65% c. 75%
2. Ava 45%, Ben 48%, Cruz 44%. Ben has the best record.
3. Rosa: 34/40 = 85/100 = 85%. Kai: 42/50 = 84/100 = 84%. Rosa did better, even though Kai spelled more words.
4. A, C (75% and 72%; 17 out of 25 = 68%, 7 out of 10 = 70% exactly)
5. Equal rates come from multiplying, not adding the same amount. 18/24 = 3/4 = 75% and 24/30 = 4/5 = 80%, so the rates are not equal; 24 out of 30 is better.

TEST PRACTICE

6. **C** (33 out of 50)
7. **B** (A won 75%, B won 60%. A did better.)

LESSON 4-5

Find the Whole · Solve for the Whole

Sheet 1 · Find the Whole

1. 20% = $18; dollars: 18, 36, 54, 72, 90. The helmet costs $90.
2. a. 130 b. 44 c. 70 d. 80 e. 140 f. 90
3. One part (25%) = 18 ÷ 3 = 6 gal; full = 4 × 6 = 24 gallons.
4. 25% = 120 ÷ 3 = 40 pages, so 100% = 4 × 40 = 160 pages.
5. A is greater: 20 is a smaller percent (40%) of A, so A must be bigger. A = 50 (10% = 5) and B = 25 (10% = 2.5, or 20 ÷ 0.8).

Sheet 2 · Solve for the Whole

1. a. 70 b. 48 c. 80
2. a. 0.55 × w = 33 b. w = 33 ÷ 0.55 = 60
3. 0.75 × w = 315, w = 315 ÷ 0.75 = $420. Still needed: $420 − $315 = $105.
4. 0.35 × w = 42, w = 42 ÷ 0.35 = 4,200 ÷ 35 = $120. Check: 0.35 × 120 = 42.
5. He divided by 35 instead of 0.35 (35% = 0.35); a whole of 0.8 is smaller than the part, which makes no sense. w = 28 ÷ 0.35 = 80. Check: 0.35 × 80 = 28.

TEST PRACTICE

6. **B** (0.3 × *w* = 45)
7. **C** (30)